№7

UNIX.

Linux, , .
, , .

5.2. 5.2.1. touch.
: 1 touch - cat. : 1 cat -
less. : 1 less -
: – Space — , – ENTER — , – b —
, – h — , – q — . head
10 . : 1 head [-n] - , n — . tail
10 . : 1 tail [-n] - , n — . . . .
47 5.2.2. cp . : 1
cp [-] _ _ : 1. . ~/abc1 april
may: 1 cd 2 touch abc1 3 cp abc1 april 4 cp abc1 may 2. .
april may monthly: 1 mkdir monthly 2 cp april may monthly 3.
. monthly/may june: 1 cp monthly/may monthly/june 2 ls
monthly i cp . ,
, cp r. : 1. . monthly
monthly.00: 1 mkdir monthly.00 2 cp -r monthly monthly.00 2. .
monthly.00 /tmp 1 cp -r monthly.00 /tmp 5.2.3.
mv mvdir . mv: 48 № 5.
Linux. ... 1 mv [-] _ _ : 1.
. april july : 1 cd 2 mv april july 2. .
july monthly.00: 1 mv july monthly.00 2 ls monthly.00 : 1 april july
june may 3. , i.
. monthly.00 monthly.01 1 mv monthly.00 monthly.01 4.

. monthly.01 reports: 1 mkdir reports 2 mv monthly.01 reports 5. , . reports/monthly.01 reports/monthly: 1 mv reports/monthly.01 reports/monthly 5.2.4. (. 5.1).
: – ((-) , (d) —); – (r— , w— , x— - ,-—); – (r— , w— , x— ,-—); – (r— , w— , x— ,-—). : . . . 49
5.1 r
w - x - (
/) - 1. (-)
(rw-), , , (r–),
(r–): 1 -rw-r–r– 2. , : 1 -rwx—— 3.
(d) , , ,
— : 1 drwxr-x–x 5.2.5. ,
chmod. ( ) . :
1 chmod _ ( ) - : = -
+ r w 50 № 5. Linux. ... x
u(user) g(group) , o(others)
( ) (. 5.2). 5.2
111 7 rwx 110 6 rw- 101 5 r-x 100 4 r– 011 3 -wx 010 2 -w- 001 1 –x 000 0—
: ~/may : 1. 1 cd 2 touch may 3 ls -l may 4 chmod u+x may 5 ls -l may 2. 1 chmod u-x may 2 ls -l may ~/may : 3.
monthly : 1 cd 2 mkdir monthly 3 chmod g-r, o-r monthly 4. 1 cd 2 touch abc1 3 chmod g+w abc1 ~/abc1
: . . . 51 5.2.6. Linux
. - . .
- : – ext2fs (second extended filesystem); – ext2fs (third extended file system); – ext4 (fourth extended file system); – ReiserFS; – xfs; – fat (file allocation table); – ntfs (new technology file system). -
mount . : 1 mount 2 10 11 12 13 14 15 3 proc on /proc type proc (rw) 4 sysfs on /sys type sysfs (rw,nosuid,nodev,noexec) 5 udev on /dev type tmpfs (rw,nosuid) 6 devpts on /dev/pts type devpts (rw,nosuid,noexec) 7 /dev/sda1 on /mnt/a type ext3 (rw,noatime) 8 /dev/sdb2 on /mnt/docs type reiserfs (rw,noatime) 9 shm on /dev/shm type tmpfs (rw,noexec,nosuid,nodev) usbfs on /proc/bus/usb type usbfs (rw,noexec,nosuid,devmode=0664,devgid=85) binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,noexec,nosuid,nodev) nfsd on /proc/fs/nfs type nfsd (rw,noexec,nosuid,nodev)
, - ( ), .
mount — ,
. /dev, (, sdaN, sdbN hdaN, hdbN, N — , sd — SCSI, hd — MFM/IDE). —
( ), .
- — /etc/fstab. cat: 1 cat /etc/fstab 2 3 /dev/hda1 / ext2 defaults 1 1 4 /dev/hda5 /home ext2 defaults 1 2 5 /dev/hda6 swap

swap defaults 0 0 6 /dev/hdc /mnt/cdrom auto umask=0,user,noauto,ro,exec,users 0 0 52 № 5. Linux. … 10 7 none /mnt/floppy supermount dev=/dev/fd0,fs=ext2:vfat,–, 8 sync,umask=0 0 0 9 none /proc proc defaults 0 0 none /dev/pts devpts mode=0622 0 0 : – ; – ; – ; – ; – dump; – fsck. - df, , . - : 1 df 2 3 Filesystem 1024-blocks Used Available Capacity Mounted on 4 /dev/hda3 297635 169499 112764 60% / fsck ( ) - : : 1 fsck _ : 1 fsck /dev/sda1

1. `abc1` `touch abc1.` `cat abc1` — , . `/etc/passwd` `less /etc/passwd.` `q.` `head -5 /etc/passwd` `tail -3 /etc/passwd.` .

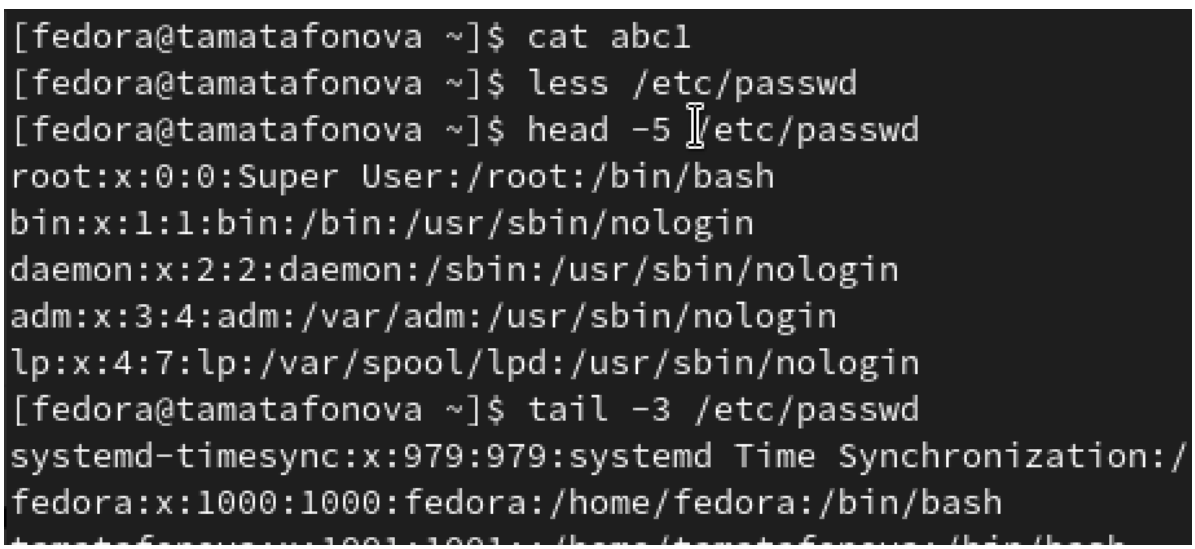


Figure 1:

2. `abc1` — `april` `may` — `cp abc1 april` `cp abc1 may.` `monthly` `mkdir monthly` : `cp april may monthly.` `may` `monthly` `june` `cp monthly/may monthly/june.` `ls monthly` `april, june, may.` : `cp`

-r monthly monthly.00 — .
/tmp cp -r monthly.00 /tmp

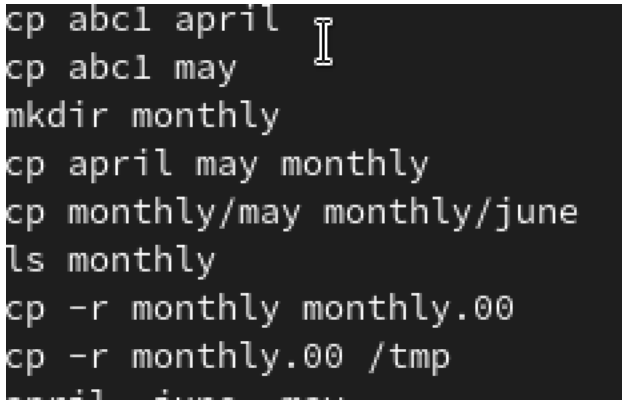


Figure 2:

3. april july mv april july. monthly.00 july mv july monthly.00. monthly.00 monthly.01 mv monthly.00 monthly.01. reports mkdir reports monthly.01 mv monthly.01 reports. monthly mv reports/monthly.01 reports/monthly. , mv , .

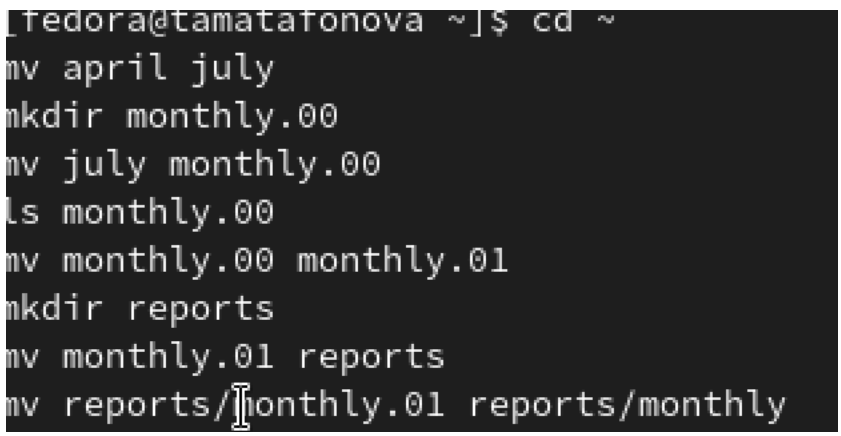


Figure 3:

4. may touch may ls -l may. chmod u+x may — x. chmod u-x may , . monthly chmod g-r,o-r monthly. abc1 chmod g+w abc1. , .

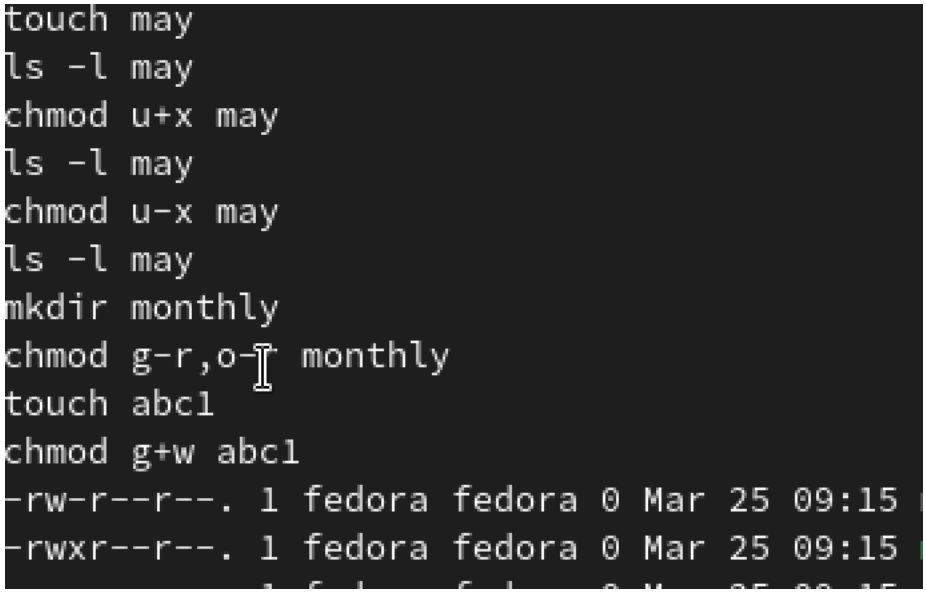
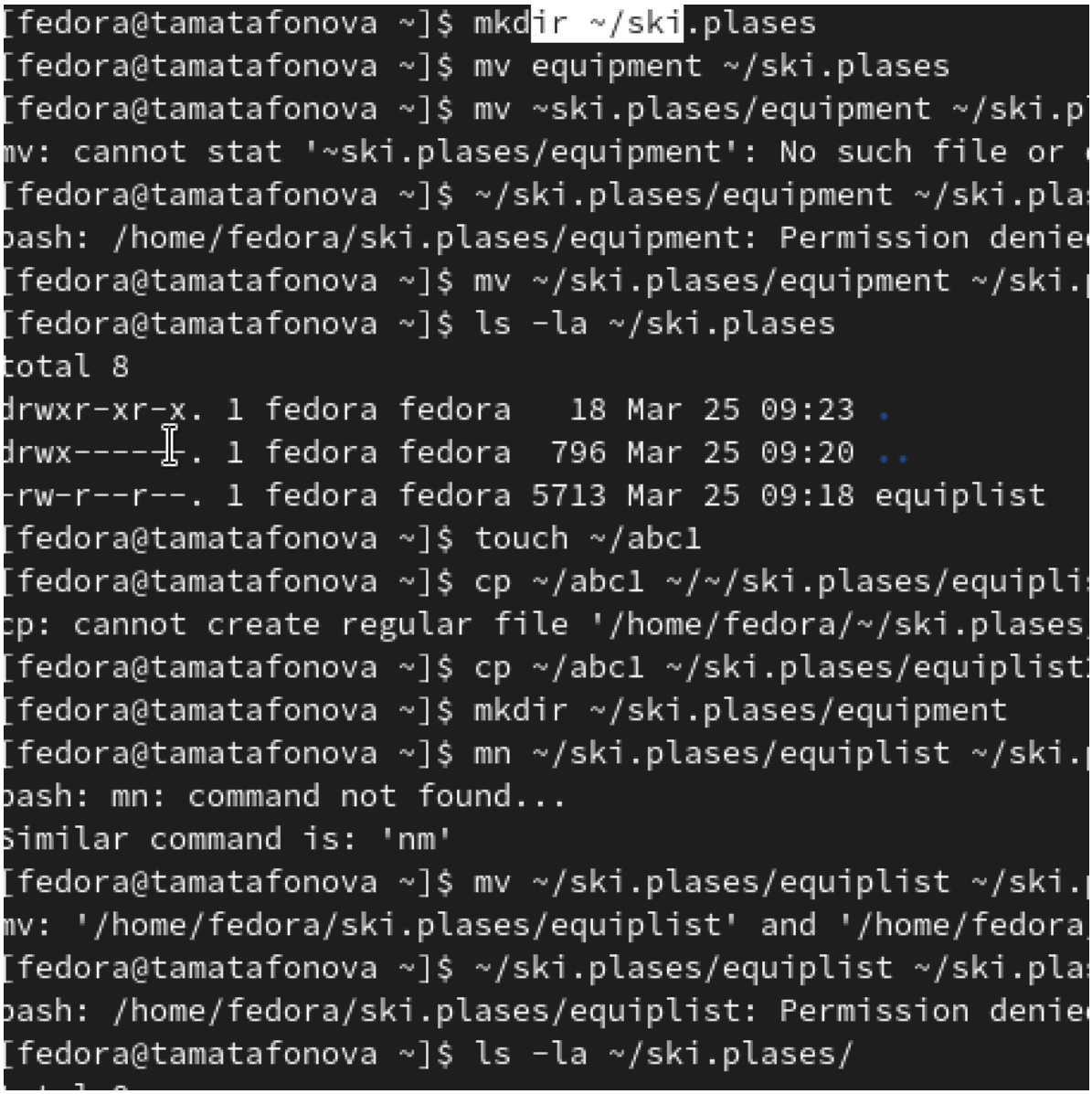


Figure 4:

5. /usr/include/sys/types.h equipment cp /usr/include/sys/types.h equipment. ~/ski.plases mkdir ~/ski.plases equipment mv equipment ~/ski.plases/. equiplist mv ~/ski.plases/equipment ~/ski.plases/equiplist. abc1 touch ~/abc1 ~/ski.plases equiplist2 cp ~/abc1 ~/ski.plases/equiplist2. ~/ski.plases equipment mkdir ~/ski.plases/equipment — equiplist equiplist2 — mv ~/ski.plases/equiplist ~/ski.plases/equiplist2 ~/ski.plases/equipment/. newdir ~/ski.plases, plans, mkdir ~/newdir mv ~/newdir ~/ski.plases/plans. ls -la ~/ski.plases/, ls -la ~/ski.plases/equipment/ ls -la ~/ski.plases/plans/ .

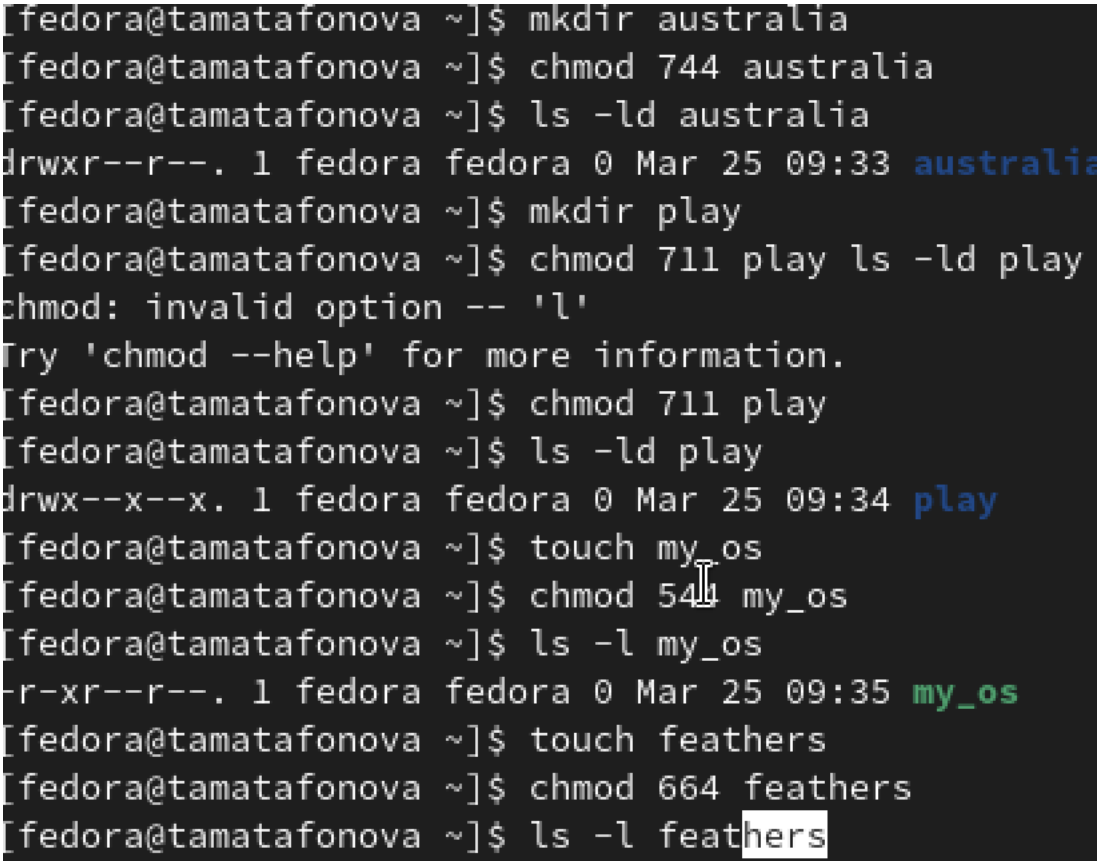
6. australia mkdir australia 744 (drwxr--r--) chmod 744 australia, ls -ld australia. play mkdir play 711 (drwx--x--x) chmod 711 play, ls -ld play. my_os touch my_os 544 (-r-xr--r--) chmod 544 my_os, ls -l my_os. feathers touch feathers 664 (-rw-rw-r--) chmod 664 feathers, ls -l feathers.

7. feathers file.old cp ~/feathers ~/file.old. file.old play mv ~/file.old ~/play/. play fun cp -r ~/play ~/fun. fun play games mv ~/fun ~/play/games. feathers chmod u-r ~/feathers



```
[fedora@tamatafonova ~]$ mkdir ~/ski.plases
[fedora@tamatafonova ~]$ mv equipment ~/ski.plases
[fedora@tamatafonova ~]$ mv ~ski.plases/equipment ~/ski.pl
mv: cannot stat '~ski.plases/equipment': No such file or
[fedora@tamatafonova ~]$ ~/ski.plases/equipment ~/ski.pla
bash: /home/fedora/ski.plases/equipment: Permission denie
[fedora@tamatafonova ~]$ mv ~/ski.plases/equipment ~/ski.
[fedora@tamatafonova ~]$ ls -la ~/ski.plases
total 8
drwxr-xr-x. 1 fedora fedora   18 Mar 25 09:23 .
drwx------. 1 fedora fedora  796 Mar 25 09:20 ..
-rw-r--r--. 1 fedora fedora 5713 Mar 25 09:18 equiplist
[fedora@tamatafonova ~]$ touch ~/abc1
[fedora@tamatafonova ~]$ cp ~/abc1 ~/~/ski.plases/equipli
cp: cannot create regular file '/home/fedora/~/ski.plases
[fedora@tamatafonova ~]$ cp ~/abc1 ~/ski.plases/equiplist
[fedora@tamatafonova ~]$ mkdir ~/ski.plases/equipment
[fedora@tamatafonova ~]$ mn ~/ski.plases/equiplist ~/ski.
bash: mn: command not found...
Similar command is: 'nm'
[fedora@tamatafonova ~]$ mv ~/ski.plases/equiplist ~/ski.
mv: '/home/fedora/ski.plases/equiplist' and '/home/fedora
[fedora@tamatafonova ~]$ ~/ski.plases/equiplist ~/ski.pla
bash: /home/fedora/ski.plases/equiplist: Permission denie
[fedora@tamatafonova ~]$ ls -la ~/ski.plases/
```

Figure 5: : ski.plases



```
[fedora@tamatafonova ~]$ mkdir australia
[fedora@tamatafonova ~]$ chmod 744 australia
[fedora@tamatafonova ~]$ ls -ld australia
drwxr--r--. 1 fedora fedora 0 Mar 25 09:33 australia
[fedora@tamatafonova ~]$ mkdir play
[fedora@tamatafonova ~]$ chmod 711 play ls -ld play
chmod: invalid option -- 'l'
Try 'chmod --help' for more information.
[fedora@tamatafonova ~]$ chmod 711 play
[fedora@tamatafonova ~]$ ls -ld play
drwx--x--x. 1 fedora fedora 0 Mar 25 09:34 play
[fedora@tamatafonova ~]$ touch my_os
[fedora@tamatafonova ~]$ chmod 544 my_os
[fedora@tamatafonova ~]$ ls -l my_os
-r-xr--r--. 1 fedora fedora 0 Mar 25 09:35 my_os
[fedora@tamatafonova ~]$ touch feathers
[fedora@tamatafonova ~]$ chmod 664 feathers
[fedora@tamatafonova ~]$ ls -l feathers
```

Figure 6:

ls -l ~/feathers. cat ~/feathers Permission denied. cp ~/feathers ~/feathers_copy . chmod u+r ~/feathers — .

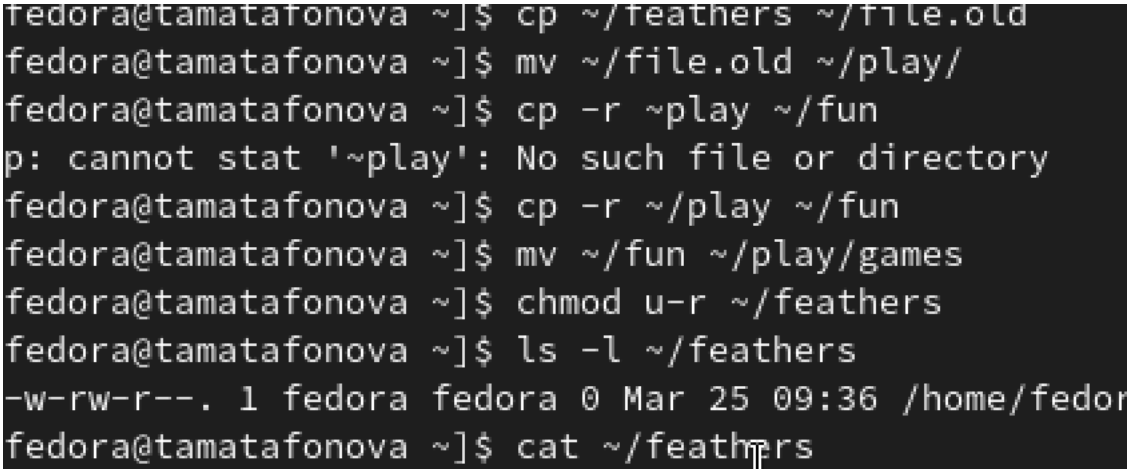


Figure 7:

8. play chmod u-x ~/play ls -ld ~/play. cd ~/play Permission denied. chmod u+x ~/play , cd ~/play pwd . mount, fsck, mkfs kill man mount | head -20, man fsck | head -20, man mkfs | head -20 man kill | head -20. .

#

1. , , : ext4 , tmpfs /dev, /dev/shm, /run, devpts , sysfs , proc . ext4 Linux, , . tmpfs , , . devpts sysfs , . proc .

2. Linux , /. : /bin — ; /boot — ; /dev — ; /etc — ; /home — ; /lib — ; /media /mnt — ; /opt — ; /proc — ; /root — ; /run — ; /sbin — ; /srv — ; /sys — ; /tmp — ; /usr — ; /var — (, . .).

```
cat ~/feathers
cp ~/feathers ~/feathers_copy
chmod u+r ~/feathers
ls -l ~/feathers
chmod u-x ~/play
ls -ld ~/play
cd ~/play
chmod u+x ~/play
ls -ld ~/play
cd ~/play
pwd
-rw-r--r--. 1 fedora fedora 0 Mar 25 09:50 /home/fedora/file.old
file.old
file.old
file.old  games
--w-rw-r--. 1 fedora fedora 0 Mar 25 09:50 /home/fedora/feathers
cat: /home/fedora/feathers: Permission denied
cp: cannot open '/home/fedora/feathers' for reading: Permission denied
-rw-rw-r--. 1 fedora fedora 0 Mar 25 09:50 /home/fedora/feathers
drw-r-xr-x. 1 fedora fedora 26 Mar 25 09:50 /home/fedora/play
bash: cd: /home/fedora/play: Permission denied
drwxr-xr-x. 1 fedora fedora 26 Mar 25 09:50 /home/fedora/play
/home/fedora/play
[fedora@tamatafonova play]$ man mount | head -20
MOUNT(8)                                  System Administration

NAME
       mount - mount a filesystem

SYNOPSIS
```

Figure 8:

3. , .
   `mount`, — .
   , `mount /dev/sdb1 /mnt/data`.
   `/etc/fstab`.

4. : , ( ),
   , , .
   `fsck` (file system check), . ,
   `fsck /dev/sda1`. , .

5. `mkfs` (make filesystem). : `mkfs -t`
   , , `mkfs -t ext4 /dev/sdb1`.
   : `mkfs.ext4`, `mkfs.xfs`, `mkfs.btrfs` .
   : , inode, . .

6. : `cat` — , ; `less` —
   , ; `head` — 10
   (`-n`); `tail` — 10 (`-n`).
   `less` , .

7. `cp`: (`cp source dest`);
   (`cp file1 file2 dir/`); (`cp -p`); (`cp -r`
   `cp -R`); (`cp -i`); (`cp -l`);
   (`cp -s`). `cp` , , .

8. `mv` . : (`mv`
   `old new`); (`mv file dir/`); (`mv file1`
   `file2 dir/`); (`mv -i`);
   . `mv` , ,
   .

9. — , , (, ,) ,
   . `chmod`. : `chmod u+x file` (
   ), `chmod g-w file` ( ), `chmod o=r file` ( ).
   : 4 — , 2 — , 1 — ; ,
   . , `chmod 755 file` `rwxr-xr-x`.

Linux
. , , `touch`,
`cp`, `mv`. , `cat`, `less`, `head`, `tail`.
Linux `chmod` , .

, , `mount`, `fsck`, `mkfs` `kill`
. , .

1. “ №7”